

Tanaya Pawar

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EDUCATION

New York University New York, USA
Masters in Computer Science, GPA: 3.88/4 Aug 2024 - May 2026
Coursework: Machine Learning, Open Source, Software Engineering, Big Data, Deep Learning, Algorithms

Savitribai Phule Pune University Pune, India
Bachelor Of Computer Engineering *with distinction*, GPA: 8.64/10 Aug 2020 - June 2024
Coursework: Data Structures, Object Oriented Programming, Operating Systems, Cloud Computing

EXPERIENCE

Referrio, New York, USA | Full Stack Developer Intern June 2025 - Aug 2025

- Developed a full-stack web app with **React**, **Next.js**, TailwindCSS & ShadCN UI improved UX & data latency
- Designed & containerized backend services using **Node.js** & **Express**, deploying microservices on **GCP**
- Built & tested **REST APIs** with postman ensuring 100% data integrity, automated CI/CD via GitHub Actions
- Created a real-time support system using **PostgreSQL** & WebSockets, cutting internal resolution time by 20%
- Delivered features in Agile sprints, accelerating product iteration in a fast-paced startup environment

New York University, New York, USA | Grader and Tutor Sep 2025 - Present

- Tutored and graded 50+ students in **Programming Languages** and **Basic Algorithms**, teaching Scala, C++, and Haskell while reinforcing algorithm design, complexity analysis, and structured problem-solving

Wilo Mather And Platt, Pune, India | Data Analyst Intern Oct 2022 - Dec 2022

- Built dashboards using **Power BI** & **SQL** to track financial risk & optimize a **\$23M** budget across departments
- Delivered monthly reports with cost alerts, helping cross-functional teams identify their high-expense areas
- Reduced expenses by **38%** in 2022 by analyzing employee behavior and operational spending patterns

PROJECTS

CrisisCast | [GitHub](#) | *Apache Kafka, Pyspark, MongoDB, Docker, FastAPI, Qdrant*

- Built real-time crisis detection system ingested 1k+ msgs/sec via Reddit, Bluesky with **Kafka**, **Spark** & **Docker**
- Served a fine-tuned flan-t5-base LLM via FastAPI to reduce false positives & improve crisis classification
- Developed fault tolerant backend with MongoDB, Qdrant, Streamlit dashboard for high risk posts in real time

Distributed Chat AI Ticketing System | [GitHub](#) | *Gemini API, FastAPI, Discord, Slack, Jira, Docker*

- In **open source** course, built a distributed system that lets users manage work tickets via chat by integrating Slack/Discord, Google Gemini, and Jira/Google Tasks into a single workflow
- Designed interface driven services using dependency injection, enabling seamless provider switching across 4+ external systems without changing core logic
- Implemented **CI/CD** with CircleCI, running unit, integration & tests against real APIs while 80%+ test coverage
- Deployed the system using infrastructure as code and production telemetry, tracking latency, success rate, and failures to diagnose real performance issues

Parameter-Efficient Transformer | [GitHub](#) | *PyTorch, Hugging Face, LoRA, Python*

- Fine-tuned **RoBERTa**-base on AG News using **LoRA**, training just 0.4% of weights for parameter efficiency
- Achieved **92.3%** test accuracy in 3 epochs, outperforming full fine-tuning with 99.6% fewer parameters
- Reduced GPU memory by **50%** and training time to under 5 min/epoch, enabling fast, scalable deployment

PUBLICATIONS

"Precision Farming Methods Using Machine Learning", Agro-Advice | *ICESTM* Feb 2024

- Conducted an analysis of 24 ML crop recommendation systems, identifying gaps in microclimate adaptability & model robustness across regions
- Proposed hybrid Random Forest & XGBoost model for precision agriculture, improved yield prediction accuracy by 17.3% including a paired cropping optimization algorithm that demonstrated 22% yield improvement

SKILLS

Programming: Python, TypeScript/JavaScript, GraphQL, C++, SQL (PostgreSQL, MySQL), NoSQL (MongoDB)
Frameworks: Django, FastAPI, React, NextJs, Nodejs, Expressjs, TailwindCSS, ShadCN UI, D3
ML/AI: PyTorch, HuggingFace Transformers, scikit-learn
APIs and Tools: REST APIs, Apache Kafka, Spark, Linux, Git, Postman
Cloud: Docker, Kubernetes, GitHub Actions, AWS (EC2, S3, RDS), GCP (Cloud Run, Cloud Storage)